

DESIGN AND ANALYSIS OF ALGORITHMS	
Course Code: AD43	Credits: 3:0:0
Pre – requisites: Basic Programming	Contact Hours: 42L
Course Coordinator: Dr. Sowmya B J	

Course Content

Unit I

Asymptotic Bounds and Representation problems of Algorithms: Computational Tractability: Some Initial Attempts at Defining Efficiency, Worst-Case Running Times and Brute-Force Search, Polynomial Time as a Definition of Efficiency, Asymptotic Order of Growth: Properties of Asymptotic Growth Rates, Asymptotic Bounds for Some Common Functions, A Survey of Common Running Times: Linear Time, $O(n \log n)$ Time, Quadratic Time, Cubic Time, $O(n^k)$ Time, Beyond Polynomial Time, Sub linear Time. Some Representative Problems, A First Problem: Stable Matching: The Problem, Designing the Algorithm, Analysing the Algorithm, Extensions, Implementing the Stable Matching Algorithm, Using Lists and Arrays: Arrays and Lists, Five Representative Problems: Interval Scheduling, Weighted Interval Scheduling, Bipartite Matching, Independent Set, Competitive Facility Location.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links:
 - <https://nptel.ac.in/courses/106101060>
 - <https://nptel.ac.in/courses/106106131>
- Impartus recording: <https://a.impartus.com/ilc/#!/course/270253/452>

Unit II

Graphs & Divide and Conquer: Graph Connectivity and Graph Traversal, Breadth-First Search: Exploring a Connected Component, Depth-First Search, Implementing Graph Traversal Using Queues and Stacks: Implementing Breadth-First Search, Implementing Depth-First Search, An Application of Breadth-First Search: The Problem, Designing the Algorithm, Directed Acyclic Graphs and Topological Ordering: The Problem, Designing and Analyzing the Algorithm, A First Recurrence: The Merge sort Algorithm: Unrolling the Merge sort Recurrence, Counting Inversions: The Problem, Designing and Analyzing the Algorithm.

Unit III

Greedy Algorithms: Interval Scheduling: The Greedy Algorithm Stays Ahead: Designing a Greedy Algorithm, Analyzing the Algorithm, Scheduling to Minimize Lateness: An Exchange Argument: The Problem, Designing the Algorithm, Optimal Caching: A More Complex Exchange Argument: The Problem, Designing and Analyzing the Algorithm, Extensions: Caching under Real Operating Conditions, Shortest Paths in a Graph: The Problem, Designing the Algorithm, Analyzing the Algorithm, The Minimum Spanning Tree Problem: The Problem, Designing Algorithms, Analyzing the Algorithms, Huffman Codes and Data Compression: The Problem, Designing the Algorithm.

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Unit IV

Dynamic Programming: Weighted Interval Scheduling: A Recursive Procedure: Designing a Recursive Algorithm, Subset Sums and Knapsacks: Adding a Variable: The Problem, Designing the Algorithm, Shortest Paths in a Graph: The Problem, Designing the Algorithm, The Maximum-Flow Problem and the Ford-Fulkerson Algorithm: The problem, Designing the Algorithm, Survey Design: The problem, Designing the Algorithm, Analyzing the Algorithm, Airline Scheduling: The problem, Designing the Algorithm, Analyzing the Algorithm.

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- Links:<https://nptel.ac.in/courses/106101060>
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- Impartus recording: <https://a.impartus.com/ilc/#/course/270253/452>

Unit V

NP and Computational Intractability: Polynomial-Time Reductions A First Reduction: Independent Set and Vertex Cover, Reducing to a More General Case: Vertex Cover to Set Cover, NP-Complete Problems: Circuit Satisfiability: A First NP-Complete Problem, General Strategy for Proving New Problems NP-Complete, Sequencing Problems: The Traveling Salesman Problem, The Hamiltonian Cycle Problem.

Suggested Learning Resources

Textbook:

1. Algorithm Design - Jon Kleinberg and Eva Tardos, Pearson, 1st Edition (2013).

Reference Book:

1. Introduction to the Design & Analysis of Algorithms - Anany Levitin, 2nd Edition, Pearson Education, 2007.

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Define the basic concepts and analyse worst-case running times of algorithms using asymptotic analysis. (PO-1,2, PSO-1,3)
2. Recognize the design techniques for graph traversal using representative algorithms. (PO-1,2,3, PSO-1,3)
3. Identify how divide and conquer works and analyse complexity of divide and conquer methods by solving recurrence. (PO-1,2,3, PSO-1,3)
4. Illustrate Greedy paradigm and Dynamic programming paradigm using representative algorithms. (PO-1,2,3,4, PSO-2,3)
5. Describe the classes P, NP, and NP-Complete and be able to prove that a certain problem is NP-Complete. (PO-1,2,3,4, PSO-2,3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	Marks	Course Outcomes (COs) addressed
Internal Test-I (CIE-I)	30	CO1, CO2, CO3
Internal Test-II (CIE-II)	30	CO4, CO5
Average of the two CIE shall be taken for 30 marks		
Other Components		
Project Based Learning	10	CO1, CO2, CO3, CO4, CO5
Report Submission	10	CO1, CO2, CO3, CO4, CO5
The Final CIE out of 50 Marks = Average of two CIE tests for 30 Marks+ Marks scored Project Based Learning +Marks scored for Report Writing		
Semester End Examination (SEE)		
Course End Examination (Answer One full question from each Unit- Internal Choice)	100	CO1, CO2, CO3, CO4, CO5